# 7.1 Angles of Polygons

Geometry 7.1 · Student sheet

Name _____ Period ____ Date _____

**I can…**

- find the sum of the interior angle measures of a polygon.
- use interior and exterior angle sums to find unknown angle measures.
- find each interior and exterior angle of a regular polygon.

**Vocabulary** · fill in the missing word

**Diagonal** — a segment that joins two _____ vertices of a polygon

**Equilateral / equiangular polygon** — all sides _____ / all angles congruent

**Regular polygon** — a convex polygon that is both _____ and equiangular

**Exterior angle** — an angle formed by one side and the _____ of the next side

## Part 1 · Discovery: triangles in polygons

Draw a convex quadrilateral, pentagon, hexagon, and heptagon. From ONE vertex, draw every diagonal. Count the triangles.

| Polygon | Number of sides | Triangles from one vertex | Sum of the interior angles |
|---|---|---|---|
| Triangle | 3 | | |
| Quadrilateral | 4 | | |
| Pentagon | 5 | | |
| Hexagon | 6 | | |
| Heptagon | 7 | | |
| n-gon | n | | |

**Exterior angles.** Draw a polygon, extend one side at each vertex, cut out the exterior angles, and put them together around a point. What do they make?

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## Part 2 · Notes: fill in the blanks

**Polygon Interior Angles Theorem:** The sum of the interior angle measures of a convex n-gon is _____.

**Corollary:** The interior angle measures of a quadrilateral add to _____°.

**Polygon Exterior Angles Theorem:** The exterior angles of a convex polygon, one at each vertex, add to _____°.

**Regular n-gon:** each interior angle = _____ each exterior angle = _____

## Part 3 · Examples (work them with the class)

### Example 1

Find the sum of the interior angle measures of a convex 12-gon.

### Example 2

The sum of the interior angle measures of a convex polygon is 1440°. Classify the polygon by its number of sides.

### Example 3

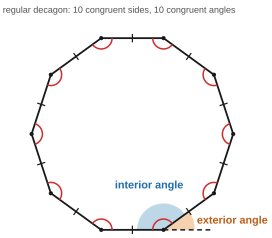
A pentagon has interior angles of 110°, 2x°, 130°, (x + 25)°, and 95°. Find x and the two unknown angles.

### Example 4

A quadrilateral's exterior angles (one at each vertex) are x°, 2x°, 89°, and 67°. Find x.

### Example 5

Find the measure of each interior angle and each exterior angle of a regular decagon.




## Part 4 · Partner practice

### Partner 1

Find the sum of the interior angle measures of a convex 15-gon.

**Partner 2**

Each exterior angle of a regular polygon measures 24°. How many sides does it have?